

Drive Three Machine Screws into a Tapped Desktop SOP (1x Set: 3 Screws)

Load each machine screw onto the electric screwdriver's magnetic bit, then drive it into a pre-tapped hole in the desktop. The screwdriver holds the screw for driving — screws are never pre-seated in the holes by hand. Drive the three holes in order **1, 2, 3** (left to right).

The table is set up in one of three ways. Only the screw holder moves; the desktop and the screwdriver cradle are in the same place in all three.

- **Config L1:** the screw holder is at the back-left, behind and left of the desktop.
- **Config L2:** the screw holder is at the front-left.
- **Config M:** the screw holder is at the front-center, in front of the desktop.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the screw holder starts at the back-left (**Config L1**), the front-left (**Config L2**) or the front-center (**Config M**). One config per episode, chosen before recording and never changed mid-episode.
- **Same-side rule:** the **left gripper** takes every screw from the holder in all three configs. The right side is not used for the holder because the right gripper holds the screwdriver for the whole set and cannot pick a screw. No arm reaches across the table.
- **Fixed roles:** the desktop is in the center, the screwdriver cradle is on the right, the **right gripper** holds the screwdriver from Step 1 to Step 5, and the holes are driven in order **1, 2, 3**.

Setup

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and the full work area is visible.
2. Both arms are home with grippers open.
3. The desktop is fixed flat and stable in the center of the work area.
4. The electric screwdriver has the correct magnetic bit and is set to forward, low speed, and the validated clutch setting.

Materials checklist

1. The desktop has three empty pre-tapped holes in a row.
2. Three matching machine screws are staged head-up in the holder in the start zone:
 - **Config L1:** back-left, behind and left of the desktop
 - **Config L2:** front-left
 - **Config M:** front-center, in front of the desktop
3. Screws, holes, and the bit are clean and undamaged; the bit holds a screw magnetically.
4. The electric screwdriver is stable in its cradle on the right.
5. The table is clear of unrelated objects.

Hole numbering

- Left to right: **1, 2, 3**

Vocabulary

- **Start zone:** where the screw holder stands at the start of the episode — back-left (**Config L1**), front-left (**Config L2**), or front-center (**Config M**). One per episode, chosen before recording and never changed mid-episode.
- **Hole order:** holes are numbered **1, 2, 3** left to right, and driven in that order. A hole is finished only when its screw is fully seated.
- **Seated on the bit:** the screw hangs from the magnetic bit on its own — straight, head held in the bit, threaded end pointing down — with the left gripper open and clear.
- **Screwdriver vertical:** the bit points straight down and the tool body looks upright, not tilted, when viewed against the desktop.
- **Fully seated head:** the screw head sits flat against the desktop with no gap under it and no visible tilt.
- **Clutch activation:** the screwdriver's clutch releases and the bit stops advancing the screw. This is the signal to release the trigger.
- **Undamaged screw:** straight shank, intact threads, and an unmarred head that the bit engages without slipping.
- **Bit empty:** no screw on the magnetic bit after the tool lifts away from a driven screw.

Steps

Drive the holes in order from hole **1** through hole **3**. Only 2.1 depends on the config: the **left gripper** takes each screw from the start zone — the back-left in Config L1, the front-left in Config L2, the front-center in Config M. Every other line is the same in all three configs.

Step 1: Pick up the screwdriver

Goal: hold the electric screwdriver vertical in the right gripper, bit empty and trigger untouched.

- With the **right gripper**, pick the screwdriver from its cradle without touching the trigger.
- With the **left gripper**, steady and seat the screwdriver in the right gripper so the bit points straight down and the screwdriver is vertical; release the left gripper once it is seated.
- Hold the screwdriver in the right gripper for the whole set. Do not re-cradle it between screws.

Step 2: Load one screw onto the bit

Goal: transfer one undamaged screw from the holder onto the magnetic bit, head seated in the bit.

2.1 Pick one screw

Look where the screw holder is before reaching for the first screw.

- **IF the holder is at the back-left (Config L1):** with the **left gripper**, grasp one screw by the sides of its head, threaded end pointing down, lift it straight up clear of the holder, and bring it forward and right toward the bit.
- **IF the holder is at the front-left (Config L2):** with the **left gripper**, grasp one screw by the sides of its head, threaded end pointing down, lift it straight up clear of the holder, and bring it right toward the bit.
- **IF the holder is at the front-center (Config M):** with the **left gripper**, grasp one screw by the sides of its head, threaded end pointing down, lift it straight up clear of the holder, and bring it back and up toward the bit, clear of the desktop.

Then, in all three:

- Confirm the screw is straight and undamaged.

2.2 Present the head to the bit

- Raise the screw so its head faces the screwdriver's magnetic bit; keep the threaded end down.
- Bring the bit and the screw head together until the magnet holds the screw and the head is seated in the bit.

2.3 Release the screw

- Open the **left gripper** only when the bit is holding the screw.

Check: the screw hangs from the bit — straight, head seated in the bit, threaded end pointing down. If it is crooked or drops, reload it once. If it still will not seat, end the episode.

Step 3: Drive the screw

Goal: drive the loaded screw straight until its head is fully seated against the desktop.

3.1 Align over the hole

- Move the screwdriver so the threaded tip enters the current hole; keep the screwdriver vertical.
- With the **left gripper**, steady the desktop at the marked holding area if it can shift.

3.2 Drive

- Press the trigger at low speed and drive the screw into the pre-tapped hole.
- Release the trigger immediately when the clutch activates.

3.3 Lift away

- Lift the screwdriver straight up so the bit separates from the seated screw head.

Check: the screw is straight, its head is fully seated against the desktop, and the head is not damaged; the bit is now empty. If the screw tilts, stops moving downward, or the bit slips, release the trigger and end the episode. Do not force the screw.

Step 4: Install the remaining screws

Goal: install one fully seated screw in each of the three holes in numerical order.

- Repeat Steps 2–3 for holes 2 and 3, keeping the screwdriver in the right gripper throughout.
- Complete one screw before loading the next screw.
- Follow hole order **1, 2, 3**.

Step 5: Finish the episode

Goal: finish with three correctly installed screws, the tool cradled, and both arms home.

- Confirm there are three straight, fully seated screws, one in each hole.
- Confirm the screw holder is empty and the bit is empty.
- Return the screwdriver to its cradle.
- Return both arms home with grippers open, then end recording.

After the episode: reset the workspace

Reset with recording off.

1. Set the screwdriver to reverse and remove screws 3 through 1.
2. Inspect the screws, holes, and bit. Replace damaged parts.
3. Return the screwdriver to forward, low speed, and the validated clutch setting.

4. Stage three screws head-up in the holder for the next episode's config — back-left (Config L1), front-left (Config L2), or front-center (Config M) — and leave all desktop holes empty.
5. Run the Setup checklist before the next episode.

SOP violations

These are actions that break the SOP and are reviewed in the side-by-side review tool.

How to record a violation in review

For every violation, record:

- The **start timestamp** of the violation in the video.
- The **violation name** from the list below.
- The **SOP rule broken**.

The visible cue is what the reviewer sees in the video. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Retain the episode in training data with the violation tag. Do not delete it solely because of a violation.

Violations

Note on the start position: the violations below were written for Config L1 (screw holder on the left of the desktop, screwdriver cradle on the right). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the screw holder is not in the start zone for the config; a gripper reaches across the table for a screw; the right gripper takes a screw from the holder; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the same-side rule (the left gripper takes every screw from the holder while the right gripper holds the screwdriver; no arm reaches across the table; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the screw holder is before the first reach, then follow that config's IF line in Step 2.1.

Violation: Wrong installation order

- **Visible cue:** a screw is driven into a higher-numbered hole while a lower-numbered hole is empty or unfinished.

- **SOP rule broken:** Steps 1–4 (drive and finish the holes in order 1, 2, 3).
- **Coaching note:** finish the current numbered hole before moving to the next one.

Violation: Screwdriver not seated squarely in the right gripper

- **Visible cue:** the screwdriver sits tilted or loose in the right gripper and the left gripper does not square it before the first screw is loaded, so the bit is not vertical when driving begins.
- **SOP rule broken:** Step 1 (use the left gripper to seat the screwdriver vertical in the right gripper before loading a screw).
- **Coaching note:** square the screwdriver in the right gripper with the left gripper before loading the first screw.

Violation: Screw not loaded on the bit before driving

- **Visible cue:** the screwdriver moves toward a hole while the left gripper still holds the screw; the screw is crooked, loose, or falling off the bit; or the bit is empty on approach.
- **SOP rule broken:** Step 2 (load the screw onto the magnetic bit, release the gripper, and confirm it is seated straight before driving).
- **Coaching note:** seat the screw on the bit and release the gripper before moving to the hole.

Violation: Misaligned screw forced

- **Visible cue:** the screw tilts, wobbles, binds, or stops moving downward and the electric screwdriver keeps running.
- **SOP rule broken:** Step 3.2 (stop immediately if the screw tilts, stops advancing, or the bit slips; never force the screw).
- **Coaching note:** release the trigger at the first sign of misalignment or binding.

Violation: Screwdriver used incorrectly

- **Visible cue:** the bit rotates before the screw tip is in the hole, the screwdriver is not vertical, the bit slips and continues rotating, or the trigger remains pressed after the clutch activates.
- **SOP rule broken:** Steps 3.1–3.2 (enter the hole with the screw tip, keep the screwdriver vertical, and release the trigger immediately when the clutch activates).
- **Coaching note:** align first, drive vertically at low speed, and stop at the clutch signal.

Violation: Incorrect final screw position

- **Visible cue:** the next screw is started while the current screw is tilted, loose, damaged, or its head is not fully seated against the desktop.
- **SOP rule broken:** Steps 3.3–4 (confirm each screw is straight, undamaged, and fully seated before loading the next screw).
- **Coaching note:** inspect and finish one screw before moving to the next hole.

Violation: Incorrect episode ending

- **Visible cue:** the episode ends without three straight and seated screws, with a screw remaining in the holder, with the screwdriver outside its cradle, with an arm away from home or a gripper closed, or before the 3-second final hold is complete.
- **SOP rule broken:** Step 5 (verify all three screws, clear the input, cradle the tool, home both arms with grippers open, and then end recording).
- **Coaching note:** complete the final verification and safe-home sequence before ending.

Non-violation failures

Failures that are not caused by how the task was run do not go in the violation set. Log them as system issues, discard the episode, and do not use them for coaching.

- **Recording or camera failure:** recording stops, frames drop, or a camera feed is lost.
- **Hardware failure:** the robot, gripper, electric screwdriver, magnetic bit, clutch, or desktop fixture fails.
- **Latent material defect:** a hidden screw or tapped-hole defect prevents correct installation.

Annotation subtasks (from SOP)

1. Pick up the electric screwdriver
2. Load one screw onto the magnetic bit
3. Align the loaded screwdriver over the hole
4. Drive and seat the screw
5. Verify the desktop and end the episode